Tableau Certified Data Analyst Master Cheat Sheet

Domain 1: Connect to and Transform Data

1.1 Connect to Data Sources

Choosing an Appropriate Data Source for Tableau Certified Data Analyst Topics

To effectively explain the various topics associated with the Tableau Certified Data Analyst exam, choosing the right data source is crucial. Here are some considerations and suggestions:

Data Relevance:

- Align the data source with the specific topics you want to explain. For example, if you're discussing data cleaning techniques, choose a dataset with inconsistencies or errors.
- Consider the audience's interests and background knowledge. If targeting beginners, simpler datasets like retail sales or movie ratings might be suitable.

Data Complexity:

- Strike a balance between simplicity and enough depth to showcase various Tableau functionalities.
- Avoid overly complex datasets that might confuse the audience or obscure learning points.
- If using a large dataset, consider filtering or sampling to focus on relevant aspects.

Accessibility and Availability:

- Choose publicly available datasets with clear usage licenses. Popular sources like Kaggle, UCI Machine Learning Repository, and government open data portals offer diverse options.
- Ensure the data format is compatible with Tableau (.csv, .xlsx, etc.).

Examples of Data Sources:

- Retail Sales: Analyze customer demographics, buying trends, and product performance.
- Movie Ratings: Explore correlations between genres, release dates, and critical reception.
- Global Temperature Data: Visualize climate trends and temperature changes across different regions.
- Website Traffic: Understand user behavior, identify popular pages, and optimize website experience.
- Sports Statistics: Analyze team performance, player comparisons, and game trends.

Live Connection:

- Benefits: Real-time updates, efficient for large datasets, avoids data duplication.
- Drawbacks: Requires constant connection to the source, slower performance with complex queries, potential security concerns.

- Use cases: When data freshness is critical, monitoring dashboards, streaming data.

Extract:

- Benefits: Offline analysis, faster performance, secure environment, portable.
- Drawbacks: Data becomes static, requires manual refresh, additional storage space.
- Use cases: Frequent analysis of large datasets, disconnected environments, sharing dashboards.

Connecting to Extracts:

- File-based sources: Connect to .tde or .hyper files created with Tableau Prep or Desktop.
- Server-based sources: Connect to extracts published on Tableau Server or Online.
- Refresh options: Automatic or manual refresh schedules, triggered by event or data changes.

Connecting to Spreadsheets:

- Excel, CSV, Google Sheets: Direct connection or import data as extract.
- Limitations: Performance issues with large spreadsheets, limited data manipulation capabilities.
- Use cases: Simple datasets, quick analysis, temporary exploration.

Connecting to .hyper Files (or .tde Files):

- High-performance data extracts: Faster loading times, efficient compression, optimized for Tableau.
- Compatibility: .hyper format is newer and preferred, .tde still supported for legacy projects.
- Creation tool: Tableau Prep or Desktop.

Choosing the Right Option:

Consider these factors:

- Data size and complexity: Choose live connection for smaller, frequently changing data, extract for large, static datasets.
- Performance needs: Live connections might be slower, extracts offer faster queries.
- Offline access: Extracts allow analysis without online connection.
- Security considerations: Live connections might expose sensitive data, consider extracts for secure environments.
- Sharing and collaboration: Extracts are portable and sharable, ideal for dashboards.

Connecting to Relational Databases:

Tableau seamlessly connects to various relational databases, allowing you to tap into diverse data repositories. Here's a breakdown of the process:

- Open Tableau Desktop: Launch the Tableau application on your computer.

- Click on "Data" > "New Data Source": This opens the data source selection window.
- Choose your database type: From the "Connect to a Server" options, select your relational database system, like MySQL, PostgreSQL, or Oracle.
- Enter connection details: Provide the server address, username, password, and database name to establish the connection.
- Explore and select tables: Browse the available tables in your database and choose the ones you want to analyze.

Pulling Data with Custom SQL Queries:

While Tableau offers a visual interface for filtering and manipulating data, sometimes, crafting custom SQL queries is necessary for precise data extraction. Here's how:

- Right-click on the chosen table: In the data source window, select the table you want to query.
- Click on "Show Data" > "Edit in Query Builder": This opens the query builder window.
- Write your SQL query: Compose your SQL query to filter, join, or aggregate data as needed. You can leverage functions, clauses, and joins to tailor your data extraction.
- Preview and confirm: Check the preview pane to ensure your query retrieves the desired data. Click "OK" to apply the query and import the data into Tableau.

Connecting to a Data Source on Tableau Server:

Tableau Server acts as a centralized platform for sharing and managing Tableau dashboards and data sources. To connect to a data source on the server:

- Open Tableau Desktop: Launch the Tableau application.
- Click on "Server" > "Sign In": Enter the Tableau Server address and your credentials to connect.
- Browse and select the data source: Navigate through the projects and folders on the server to locate the desired data source.
- Connect and explore: Click on the data source to establish the connection and start analyzing the data in Tableau.

Replacing Data Sources in Existing Charts:

Tableau allows you to dynamically replace the data source behind an existing chart or sheet, maintaining the visualization layout while updating the underlying data. Here's how:

- Select the chart or sheet: Click on the chart or sheet you want to update.
- Go to "Data" > "Replace Data Source": This opens the data source selection window.
- Choose the new data source: Select the new data source you want to connect to your chart.
- Map fields: Tableau automatically attempts to map fields from the new data source to the existing chart. You can fine-tune the mapping to ensure accurate data representation.

- Refresh and analyze: Click "OK" to replace the data source and refresh the chart. Explore the updated visualization and gain insights from the new data.

1.2. Prepare data for analysis

Assessing Data Quality:

- Completeness: Ensure that all necessary data points are present without missing values. Check for blank fields, null values, or incomplete records.
- Consistency: Verify that data is consistent across different sources or fields. Look for inconsistencies in formatting, naming conventions, or values.
- Accuracy: Validate that data is accurate and represents real-world values. Identify errors, typos, or incorrect data entry.

Performing Cleaning Operations:

- Fixing errors: Correct typos, inconsistencies, or missing values.
- Filtering: Remove unwanted or irrelevant data.
- Normalizing: Ensure consistent formatting and date/time representations.
- Aggregating: Combine data into summary statistics.

Organizing Data into Folders:

- Create a logical structure for data sources and workbooks within Tableau.
- Improve navigation and maintainability.
- Facilitate collaboration and sharing.

Using Multiple Data Sources:

- Relationships: Establish connections between tables based on shared fields.
- Joins: Combine data from multiple sources based on common fields.
- Unions: Merge data from similar tables into a single table.
- Blending: Combine data from different sources without a common field, using dimensionality.

Preparing Data with Data Interpreter, Pivot, and Split:

- Data Interpreter: Automatically cleans and prepares data for analysis.
- Pivot: Reshapes data by rotating rows into columns or vice versa.
- Split: Divides data into multiple fields based on a delimiter.

Creating Extract Filters:

- Reduce data size and improve performance by filtering data at the extract level.
- Apply filters before the data is brought into Tableau.

Mastering these skills is crucial for Tableau Certified Data Analysts to:

- Ensure the quality and reliability of their analyses.
- Effectively manage and organize data sources.
- Combine and manipulate data from multiple sources.
- Prepare data for analysis efficiently.
- Optimize performance and create accurate visualizations.

1.3. Perform data transformation in Tableau Prep

Choosing Data Transformations Based on Business Scenarios:

To effectively choose data transformations, consider these factors:

- Business Problem: Clearly understand the business question or problem you're trying to address with the data. This guides the types of transformations needed.
- Data Quality: Assess the data for issues like missing values, outliers, inconsistencies, and incorrect formats. Different transformations address different quality issues.
- Transformation Techniques: Tableau offers various tools for data transformation, including:
    - Filtering: Subset data based on specific criteria.
    - Aggregation: Summarize data by groups or categories.
    - Joins: Combine data from multiple tables.
    - Calculations: Create new fields or modify existing ones using formulas.
    - Data Cleaning: Correct errors, format values, and handle missing data.

Choosing the right transformation requires understanding the impact it has on the data and how it affects the final analysis. For example, filtering out certain data points might bias the results, while aggregating data might lose granularity.

Here are some common business scenarios and suitable data transformations:

- Scenario: Analyze sales trends across different regions.
    - Transformations: Filter data by region, aggregate sales by month, calculate average sales per region.
- Scenario: Identify factors influencing customer churn.
    - Transformations: Join customer data with purchase history, create calculated fields for customer lifetime value and churn rate, filter by churned customers, group by factors like demographics and purchase behavior.

Combine Data by Using Unions:

- Scenario: When you have similar datasets from different sources that need to be analyzed together, but they don't have any matching fields to join on.
- Example: Combining sales data from different regions into a single dataset for trend analysis.

Combine Data by Using Joins:

- Scenario: When you have related datasets with matching fields, but you need to combine them based on those fields to uncover relationships and dependencies.
- Example: Joining customer data with their purchase history to analyze buying patterns.

Shape Data by Using Aggregations:

- Scenario: When you need to summarize large datasets by grouping and calculating average, sum, count, or other statistical measures.
- Example: Calculating average monthly sales figures for each product category.

Perform Filtering:

- Scenario: When you only want to analyze a subset of your data based on specific criteria.
- Example: Filtering out sales data from a specific region or focusing on transactions above a certain value.

Shape Data by Using Pivots:

- Scenario: When you want to change the way your data is organized by rearranging rows and columns to reveal different insights.
- Example: Pivoting a dataset with customers in rows and products in columns to analyze individual customer purchase behavior.

Explain the above topics in detail:

- Unions: Imagine two separate tables of fruits, one from a grocery store and another from a farmer's market. Unions combine these tables side-by-side, keeping all rows and columns intact.
- Joins: Think of connecting two tables by a common field, like "fruit_name." Joins merge rows based on this link, allowing you to analyze relationships between data points.
- Aggregations: Picture a large bucket of apples. Aggregations summarize them by counting, averaging, or summing specific qualities, like weight or variety.
- Filtering: Imagine sifting through the apple bucket, keeping only red apples. Filtering lets you isolate specific data points based on your criteria.
- Pivots: Think of rotating the apple bucket on its side. Pivots reorganize data, showcasing different relationships, like grouping apples by variety instead of origin.

1.4. Customize fields

Change default field properties (types, sorting, etc.)

- Data types: Tableau automatically assigns data types to fields when you connect to a data source. However, you may need to change these types:
    - Convert text to dates for accurate time-based analysis.

    - Convert numeric fields to categories for grouping and sorting.
- Sorting: Control the order of values in visualizations:
    - Sort ascending or descending.
    - Apply custom sorts for specific fields.
- Other properties: Customize fields like:
    - Number formatting (decimal places, currency, etc.).
    - Date formatting (display as year, month, day, etc.).

Rename columns

- Rename fields for clarity and consistency:
    - Use descriptive names that align with analysis goals.
    - Improve readability and comprehension of visualizations.

Choose when to convert between discrete and continuous

- Discrete (blue pills): Represent distinct categories or values (e.g., city, product type).
- Continuous (green pills): Represent numerical values that can be measured along a scale (e.g., sales, temperature).
- Conversion: Necessary for certain calculations and visualizations:
    - Calculate averages or sums for continuous fields.
    - Create histograms or box plots for continuous fields.

Choose when to convert between dimension and measure

- Dimensions (blue pills): Define the structure of the data (e.g., customer names, dates).
- Measures (green pills): Represent quantifiable values used for calculations (e.g., sales amount, profit).
- Conversion: Allows different analysis techniques:
    - Create aggregated measures like sums or averages.
    - Use dimensions for grouping and filtering.

Create aliases

- Assign alternative names to fields for:
    - User-friendly display in visualizations.
    - Custom labels in tooltips or legends.
    - Creating calculated fields that reference multiple fields.

Domain 2: Explore and Analyze Data

2.1. Create calculated fields

Write date calculations (DATEPARSE, DATENAME…)

Tableau offers a powerful arsenal of date functions to manipulate and analyze temporal data. Here are some key examples:

- DATEPARSE: Converts text strings into specific date formats. E.g., DATEPARSE("DD/MM/YYYY", "25/01/2024") returns the date January 25, 2024.
- DATENAME: Extracts specific components from a date. E.g., DATENAME('month', [Order Date]) returns the month of each order date.
- DATEDIFF: Calculates the difference between two dates in units like days, years, or months. E.g., DATEDIFF('days', [Start Date], [End Date]) calculates the duration of a project.
- DATEADD: Adds or subtracts specific units to a date. E.g., DATEADD('month', 2, [Order Date]) finds the date two months after each order.

Write string functions

String functions manipulate text data to extract, clean, and format information. Some crucial ones include:

- LEFT/RIGHT/MID: Extract specific characters from the beginning, end, or middle of a string. E.g., LEFT([Customer Name], 5) returns the first five characters of a name.
- TRIM/LOWER/UPPER: Remove whitespace, convert all characters to lowercase or uppercase. E.g., TRIM([Product Description]) cleans leading and trailing spaces.
- INSTR/FIND: Find the position of a substring within a string. E.g., INSTR([Product Description], "discount") identifies the location of "discount" in the description.
- REPLACE/SUBSTITUTE: Replace specific characters or substrings with new values. E.g., REPLACE([Address], " ", "-") replaces spaces with hyphens in addresses.

Write logical and Boolean expressions (If, case, nested, etc.)

Logical and Boolean expressions control the flow of calculations based on conditions. Mastering these is key for creating dynamic and insightful visualizations.

- IF/THEN/ELSE: Perform calculations based on conditions. E.g., IF [Profit] > 100 THEN "High Profit" ELSE "Low Profit" classifies profits into categories.
- CASE: Simplify nested IF statements with multiple conditions. E.g., CASE [Region] WHEN "North" THEN "Cold" WHEN "South" THEN "Hot" ELSE "Moderate" END assigns climate labels based on regions.
- NOT/AND/OR: Combine conditions for more complex logic. E.g., NOT [Discount Applied] AND [Price] > 1000 identifies high-priced non-discounted products.

Write number functions

Number functions perform calculations on numeric data to summarize and analyze financial metrics, sales figures, and other quantitative information.

- SUM/AVG/MIN/MAX: Calculate basic statistics like total, average, minimum, and maximum values. E.g., SUM([Sales Amount]) finds the total sales for a specific period.
- MEDIAN/PERCENTILE: Determine the middle value and distribution percentiles within a dataset. E.g., MEDIAN([Order Value]) identifies the typical order size.
- ROUND/FLOOR/CEIL: Round numbers to specific decimals, round down, or round up. E.g., ROUND([Average Rating], 2) displays average ratings with two decimal places.
- ABS/SQRT/EXP: Perform absolute values, square roots, and exponential calculations for advanced analysis. E.g., ABS([Profit Margin]) analyzes the magnitude of profit margins without considering positive or negative values.

Type Conversion Functions:

These functions transform data of one type into another. Understanding them allows you to analyze data regardless of its original format. Here are some key functions:

- INT() and FLOAT(): Convert data to integers and decimals, respectively. Useful for numerical calculations.
- STRING() and STRREPLACE(): Convert data to text and manipulate text values through replacements.
- DATE() and DATETIME(): Convert data to date or datetime formats for temporal analysis.
- IFNULL() and NULLIF(): Handle null values by replacing them with specific values or removing rows containing them.

Aggregate Functions:

These functions summarize data across groups or dimensions. They provide essential insights into overall trends and patterns. Here are some common functions:

- SUM(), AVG(), MIN(), MAX(): Calculate basic aggregations like sum, average, minimum, and maximum values.
- COUNT(), COUNTD(): Count distinct or total occurrences of a dimension.
- MEDIAN(), QUARTILE(): Find the middle value or specific quartiles in a distribution.
- STDEV(), VAR(): Calculate standard deviation and variance to measure data spread.

FIXED LOD Calculations:

Level of Detail (LOD) calculations allow you to aggregate data at different levels within a hierarchy. FIXED LOD lets you fix the aggregation level regardless of filters or dimensions in the view. This allows for consistent comparisons across different contexts. Here's an example:

- Calculate year-over-year sales growth, where the sales are aggregated at the product level but compared at the overall year level regardless of filters on specific products. This provides a consistent view of growth across all products.

Explanations:

Understanding these functions and calculations enhances your data analysis capabilities. Let's delve deeper into their explanations:

- Type conversion functions: Imagine you have data with mixed types like text and numbers. Converting them allows for calculations like calculating average sales from price data stored as text.
- Aggregate functions: These functions condense large datasets into key metrics. For example, calculating average customer orders across different regions gives you a quick overview of regional performance.
- FIXED LOD calculations: Think of it as zooming in on specific data points while maintaining context. For instance, analyzing product performance within a specific region while comparing it to the overall national performance requires FIXED LOD calculations.

2.2. Create quick table calculations

Moving average

A moving average is a calculation that takes the average of a set of data points over a certain period of time. This can be helpful for smoothing out fluctuations in the data and identifying trends. In Tableau, you can calculate a moving average using the AVG function and the WINDOW clause.

Percent of total

The percent of total is a calculation that shows the percentage of each data point relative to the sum of all data points. This can be helpful for understanding the relative size of different data values. In Tableau, you can calculate the percent of total using the PERCENTILE function.

Running total

A running total is a calculation that shows the sum of all data points up to a certain point. This can be helpful for tracking trends over time. In Tableau, you can calculate a running total using the RUNNING_TOTAL function.

Difference and percent of difference

The difference is the calculation of the change between two data points. The percent of difference is the difference expressed as a percentage of the first data point. These calculations can be helpful for understanding how data is changing over time. In Tableau, you can calculate the difference and percent of difference using the DIFF and PCTDIFF functions, respectively.

Percentile

A percentile is a value that divides a set of data into 100 equal parts. For example, the 25th percentile is the value that is below 25% of the data points and above 75% of the data points. Percentiles can be helpful for understanding the distribution of data. In Tableau, you can calculate percentiles using the PERCENTILE function.

Compound growth rate

The compound growth rate (CAGR) is a calculation that shows the average annual rate of growth of a data set over a period of time. The CAGR is calculated by taking the nth root of the ratio of the ending value to the beginning value, where n is the number of years in the period. In Tableau, you can calculate the CAGR using the CEILING and LN functions.

2.3. Create custom table calculations

Year to date (YTD):

- Calculates the sum of a measure from the beginning of the current year to the present date.
- Useful for tracking progress towards annual goals or identifying seasonal trends.
- Example: YTD sales compared to target, YTD website traffic by month.

Month to date (MTD):

- Calculates the sum of a measure from the beginning of the current month to the present date.
- Helps monitor short-term performance and compare it to historical averages.
- Example: MTD revenue compared to previous months, MTD customer acquisition rate.

Year over Year (YoY):

- Compares the value of a measure at the same point in time (e.g., January 1st) over two consecutive years.
- Shows change or growth over time.
- Example: YoY sales growth, YoY website traffic change.

Index:

- Converts a series of values into a relative scale (often with a base of 100) to facilitate comparison over time or between different groups.
- Helps identify high and low performers or track deviations from the average.
- Example: Sales index per region, customer satisfaction index over time.

Ranking:

- Orders data points based on a chosen measure, assigning them numeric ranks (e.g., 1st, 2nd, 3rd).
- Provides quick insights into relative performance or distribution.
- Example: Ranking customers by revenue, ranking products by market share.

First-last:

- Calculates the difference between the first and last values of a measure in a set or table.
- Shows the overall change or movement within that timeframe.
- Example: First-last change in stock price, first-last difference in website visits.

2.4. Create and use filters

Apply filters to dimensions and measures

- Dimensions: Categorical fields that describe your data (e.g., customer names, product categories, regions).
- Measures: Numerical fields that quantify your data (e.g., sales, profits, counts).
- Filtering: Refining the data displayed in a visualization by applying conditions.

Steps to apply filters:

1. Drag a dimension or measure from the Data pane to the Filters shelf.
2. Choose filter types (e.g., General, Wildcard, Condition).
3. Select values to include or exclude.

Configure filter settings

- Top N and Bottom N: Display the highest or lowest N values based on a measure.
- Include and Exclude: Specify values to keep or remove.
- Wildcard: Use patterns to match text values (e.g., *son to find "Johnson").
- Conditional: Create complex filters using calculated fields or logical expressions.

Add filters to context

- Context filters: Applied to all sheets using the same data source, persisting across views.
- Global filters: Applied to all sheets and data sources in a workbook.

Apply filters to multiple sheets and data sources

- Apply filters to multiple sheets: Right-click a filter and choose "Apply to Worksheets" or "Apply to Selected Worksheets."
- Apply filters to multiple data sources: Use blended data sources or data relationships.

Key points:

- Filters narrow down the data to focus on specific insights.
- Choose appropriate filter types based on data types and analysis goals.
- Manage filter scope and impact using context and global filtering.
- Apply filters consistently across multiple sheets and data sources.

2.5. Create parameters to enable interactivity

Calculations

Calculations extend the capabilities of your data by creating new fields or modifying existing ones. Tableau offers a wide range of calculation types, including:

- Basic calculations: These involve arithmetic operations like sum, average, median, and standard deviation.
- String calculations: Manipulate text data using functions like UPPER, LOWER, and CONTAINS.

- Date and time calculations: Extract specific components from dates and times or perform calculations on them.
- Logical calculations: Use IF/THEN statements to define conditions and create new values based on them.
- Custom calculations: Write custom formulas using Tableau's formula language to perform complex calculations.

Here's an example of using a calculation to create a new field:

Imagine you have a dataset with sales figures and want to calculate the percentage change in sales from the previous year. You can create a new calculated field with the following formula:

(SUM([Sales]) - SUM([Sales Previous Year])) / SUM([Sales Previous Year]) * 100

This formula calculates the difference between sales this year and the previous year, then divides that difference by the previous year's sales and multiplies by 100 to express the result as a percentage.

Filters

Filters allow you to focus your analysis on specific subsets of your data. You can filter based on dimensions, measures, or calculated fields. Tableau offers various filter types, including:

- Quick filters: Provide a quick way to filter data by selecting values from a list.
- Conditional filters: Filter based on custom conditions you define using formulas.
- Context filters: Apply filters from one visualization to other visualizations on the same dashboard.

Here's an example of using a filter to explore trends:

Imagine you're analyzing sales data by region. You can use a quick filter to select specific regions and see how sales trends differ among them. This allows you to compare performance across different markets and identify areas needing attention.

Reference Lines

Reference lines help you interpret your data by adding context and visual cues. They can be horizontal or vertical lines representing specific values, ranges, or goals. Common use cases for reference lines include:

- Highlighting targets or thresholds: Draw a horizontal line at your sales target to see how close you are to achieving it.
- Comparing data points to averages or medians: Add a vertical line at the average sales value to see how individual regions perform relative to the average.
- Visualizing trends over time: Plot a trend line to show how a measure changes over time and identify periods of growth or decline.

2.6. Structure the data

Sets:

Sets in Tableau allow you to create custom categories of data points based on specific criteria. These sets dynamically update as your data changes, offering flexibility in your analysis. Here's what you can do with sets:

- Filter data: Create sets to highlight specific segments of your data, like high-performing regions or underperforming product categories.
- Color-code data points: Visualize differences within your data by assigning unique colors to different sets.
- Calculate aggregate measures: Analyze specific subsets of your data by applying calculations like averages or sums to sets.
- Create calculated fields: Use set membership to create new fields based on set criteria, enriching your analysis.

Bins:

Bins group continuous data into discrete ranges, making it easier to analyze trends and patterns. You can define the size and number of bins, or let Tableau automatically determine them based on your data distribution. Bins are helpful for:

- Summarizing large datasets: Quickly understand the overall data distribution without getting overwhelmed by individual data points.
- Identifying trends and outliers: Spot spikes, dips, and unusual values within your data by analyzing binned values.
- Creating histograms and bar charts: Visualize the distribution of your data using bins as the basis for your chart's bars.
- Comparing different datasets: Analyze two or more datasets by binning them consistently, revealing similarities and differences.

Hierarchies:

Hierarchies organize data into parent-child relationships, showing how different levels of detail relate to each other. This is particularly useful for analyzing data with inherent hierarchies, like organizational structures, product categories, or geographical regions. Hierarchies allow you to:

- Drill down into data: Analyze progressively finer levels of detail within your hierarchy, starting from a broad overview and zooming in to specific subcategories.
- Create multi-level charts: Visualize hierarchies using treemaps, pie charts with nested levels, or other chart types that showcase parent-child relationships.
- Calculate aggregate measures at different levels: Analyze data at different levels of your hierarchy, like calculating sales by region and then drilling down to see sales within each city.
- Filter data based on hierarchy levels: Focus your analysis on specific portions of your hierarchy, like analyzing sales for only a particular product category.

Groups:

Groups provide another way to categorize your data, similar to sets, but with some key differences. Groups are static, defined when you create them, and don't update automatically like sets. They are particularly useful for:

- Organizing complex data: Categorize data points based on multiple conditions or criteria, creating custom groupings that fit your specific analysis needs.
- Filtering data based on groups: Easily isolate specific groups of data for analysis, like filtering on customer segments or product types.
- Creating calculated fields based on groups: Similar to sets, you can use group membership to create new fields that enrich your analysis.
- Comparing performance across groups: Analyze different groups of data side-by-side, highlighting similarities and differences in their performance or characteristics.

2.7. Map data geographically

Symbol Maps:

- Use case: Represent individual data points on a map, such as store locations, crime incidents, or disease outbreaks.
- Strengths: Effective for visualizing large datasets with distinct locations, highlighting clusters or patterns.
- Example:



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Tableau Symbol Map showing Starbucks locations in New York City

Heat Maps:

- Use case: Depict the intensity of a measure across geographic areas, using color gradients to represent variation.
- Strengths: Quickly identify areas with high or low values, revealing trends and hotspots.
- Example:

Density Maps:

- Use case: Show the concentration of points within a specific area, often used for population density or traffic flow.

- Strengths: Effective for visualizing data where points may overlap, preventing obscuring important information.
- Example:

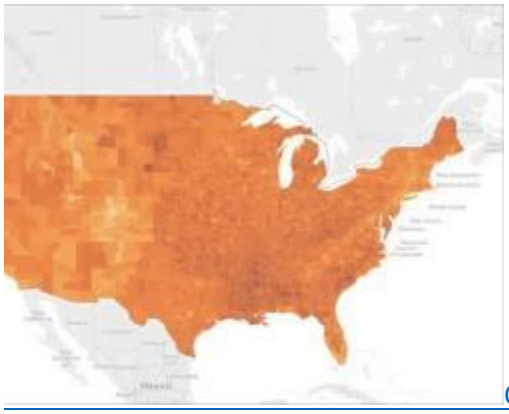


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Tableau Density Map showing population density in Los Angeles County

Choropleth Maps (Filled Maps):

- Use case: Color-code geographic regions based on their aggregate values, such as average income, election results, or crime rates.
- Strengths: Provide a quick overview of how a measure varies across regions, facilitating comparisons.
- Example:



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Tableau Choropleth Map showing unemployment rates by state in the United States

Bonus Tip:

- Tableau also offers other map types like treemaps and wind maps, catering to specific data and visualization needs.

2.8. Summarize, model, and customize data by using the Analytics feature

Totals and Subtotals:

Totals and subtotals provide summary statistics for your data.

- Totals: Represent the aggregated value for an entire data set. For example, if you're analyzing sales data, the total might be the overall sales figure.
- Subtotals: Group data into smaller categories and display the aggregated value for each group. Imagine analyzing sales by region; subtotals would show the total sales for each region.

Tableau offers various ways to display totals and subtotals:

- Shelf Marks: Drag the measure you want to summarize to the "Marks" shelf. Tableau automatically displays the total for that measure.
- Quick Calculations: Use quick calculations like "SUM" or "AVG" to calculate specific totals or subtotals.
- Table Calculations: Perform more complex calculations like running totals or grand totals.

Reference Lines:

Reference lines act as benchmarks for comparison within your data. They can be horizontal or vertical lines drawn at specific values to highlight important thresholds or targets.

- Horizontal Lines: Often used to represent averages, medians, or goals. For instance, in a sales analysis, you might draw a horizontal line at the average sales figure to see how individual regions perform.
- Vertical Lines: Typically used to mark specific dates or events. Imagine analyzing website traffic; you could draw a vertical line on the date of a marketing campaign launch to assess its impact.

Tableau allows you to customize reference lines with various options:

- Line Style and Color: Change the thickness, style, and color of the line for better visibility.
- Labeling: Add labels to explain the purpose of the reference line.
- Interactive Features: Make lines clickable to trigger actions like filtering data or displaying tooltips.

Reference Bands:

Similar to reference lines, reference bands highlight specific ranges within your data. They are shaded areas between two horizontal lines, visually emphasizing thresholds or deviations from the norm.

- Highlighting Variability: Use reference bands to show standard deviations or quartiles, providing context for data distribution.
- Tracking Targets: Define performance targets as reference bands to monitor progress towards goals. For example, in a customer satisfaction survey, bands could represent acceptable, good, and excellent satisfaction levels.

Tableau offers similar customization options for reference bands as for lines.

Average Lines:

Average lines, as the name suggests, display the average value of a measure across your data. They can be a helpful way to identify trends and outliers.

- Understanding Central Tendency: Visualize the average performance across different categories or over time.
- Identifying Deviations: Compare individual data points to the average line to spot anomalies or significant fluctuations.

Tableau lets you configure average lines with various adjustments:

- Line Style and Color: Differentiate the average line from other elements in your visualization.
- Shading: Apply shading above or below the line to emphasize the area above or below average.
- Conditional Formatting: Change the line style or color based on specific conditions.

Trend Lines:

Trend lines forecast future values based on historical data. They can be linear, exponential, or logarithmic, depending on the underlying trend in your data.

- Predicting Future Outcomes: Use trend lines to estimate future performance or resource requirements.
- Understanding Growth Patterns: Visualize how a measure is changing over time and anticipate future trends.

Tableau offers different trend line options:

- Trend Line Model: Choose the appropriate model (linear, exponential, etc.) based on your data.
- Confidence Intervals: Show the range within which future values are likely to fall, indicating the uncertainty of the prediction.
- Forecasting Duration: Specify how far into the future you want the trend line to extend.

Distribution Bands:

- These are visual elements overlaid on a chart to show the spread of data around a chosen metric (e.g., average).
- Tableau offers several types of distribution bands, including standard deviation, confidence intervals, and quantiles.
- Data analysts use them to:
    - Understand the variability of data within a group.
    - Assess the significance of differences between groups.
    - Identify outliers.

Forecast by using default settings:

- Tableau has built-in forecasting capabilities using a statistical model called Exponential Smoothing.

- By simply dragging a time series field onto the "Marks" shelf, you can automatically generate a basic forecast.
- This is helpful for quick insights, but analysts can:
    - Customize the forecast: Choose different smoothing factors, adjust seasonality settings, and set the forecast horizon.
    - Evaluate the forecast: Analyze the model's fit and accuracy using error metrics like Mean Absolute Error (MAE).

Customize a data forecasting model:

- Tableau allows manual model selection from various options like ARIMA, Holt-Winters, and Prophet.
- Analysts can fine-tune parameters, define trends, and incorporate external factors like holidays or promotions.
- This requires deeper understanding of forecasting models and data analysis principles.

Create a predictive model:

- While Tableau is primarily for data visualization, you can build basic predictive models using calculated fields and regressions.
- This involves identifying independent variables (predictors) and using them to estimate a dependent variable (target).
- Predictive models are powerful for:
    - Identifying key drivers of business outcomes.
    - Building dashboards with predictive indicators.
    - Automating workflows with insights from the model.

To effectively utilize these techniques, a Tableau Certified Data Analyst should:

- Understand the strengths and limitations of each forecasting/predictive approach.
- Interpret the results and communicate insights effectively to stakeholders.
- Continuously monitor and validate models to ensure their relevance and accuracy.

Additional Resources:

- Tableau Forecasting Guide: https://m.youtube.com/watch?v=ZB78DglzyZY
- Building Predictive Models in Tableau: https://help.tableau.com/current/pro/desktop/en-us/predictions.htm
- Tableau Desktop Specialist Exam Guide: https://www.tableau.com/learn/certification/tableau-desktop-specialist-exam-guide

Domain 3: Create Content

3.1. Create charts

Basic Sorting:

- Sort by Dimension: Arrange data alphabetically, numerically, or by date within a single dimension. For example, sorting customers by their names or purchases by date of sale.
- Sort by Measure: Order data based on the values of a measure. For example, sorting products by their sales volume or customer segments by their average order value.

Custom Sorting:

- Sort by Expression: Create custom formulas and expressions to define unique sorting criteria. This allows you to sort by complex calculations, ratios, or combinations of dimensions and measures.
- Custom Sort Order: Set custom sort orders for specific dimensions, overriding the default alphabetical or numerical ordering. This is useful for categorizing data into specific groups or hierarchies.

Nested Sorting:

- Sort by Multiple Dimensions: Apply multiple sorting levels, ordering data by one dimension and then further refining the order within each group based on another dimension. For example, sorting customers by region and then by sales within each region.

Sorting in Charts and Views:

- Drag-and-Drop Sorting: Directly manipulate the sort order of dimensions and measures by dragging and dropping them in the "Marks" card or the "Columns" and "Rows" shelves.
- Toolbar Sorting: Use the sort options in the toolbar to quickly sort any element on the chart by ascending or descending order.

Understanding Sorting for the Exam:

While the Tableau Certified Data Analyst exam won't directly ask you to perform specific sorts, understanding these techniques is crucial for several exam topics:

- Data Analysis and Exploration: Sort data to identify trends, patterns, and outliers within your dataset.
- Building Effective Visualizations: Choose appropriate sorting methods to present data in a clear and concise manner, supporting your analysis and conclusions.
- Problem-Solving Scenarios: Apply sorting techniques to answer specific business questions and solve data-driven problems.

Tips for Mastering Sorting:

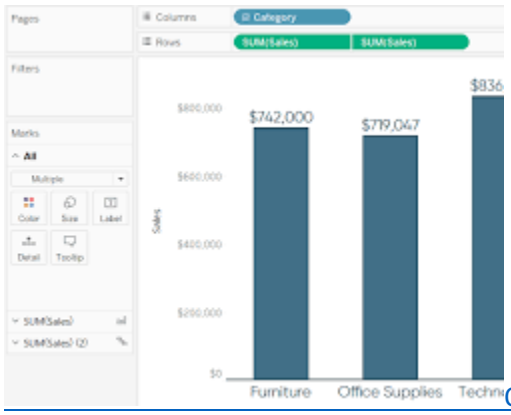
- Practice sorting data in Tableau Desktop with various datasets and visualizations.
- Experiment with custom expressions and sort orders to create unique analysis perspectives.

- Understand the context of your analysis and choose the sorting method that best supports your goals.
- Utilize Tableau resources and training materials to deepen your understanding of sorting techniques.

Creating Basic Charts:

Bar Chart: Ideal for comparing categories or showing trends over time.

- Drag and drop your measure (e.g., sales) onto the "Marks" shelf and dimension (e.g., product) onto the "Columns" shelf.
- You can customize the chart by changing the chart type (e.g., stacked bar), adding color, or applying labels.



Opens in a new window playfairdata.com

Bar Chart in Tableau

Line Chart: Useful for showing trends over time or comparing multiple measures.

- Drag your measure onto the "Marks" shelf and dimension onto the "Columns" shelf.
- You can add multiple measures to create a line chart with multiple lines.



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Line Chart in Tableau

Pie Chart: Effective for showing proportional relationships between parts of a whole.

- Drag your dimension onto the "Marks" shelf and measure onto the "Angle" shelf.

- You can customize the chart by adding labels, adjusting slice sizes, and applying colors.



Opens in a new window btprovider.com

Pie Chart in Tableau

Highlight Table: Combines a table with a chart to emphasize specific data points.

- Create your desired chart (e.g., bar chart).
- Drag another dimension onto the "Text" shelf to create a table.
- Use the "Marks" card to format the table and highlight specific data points.



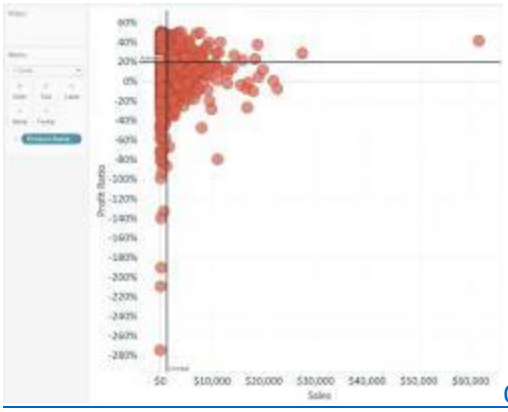
	January	February	March	Ap
Accessories	$5,478	$5,369	$8,735	
Appliances	$3,176	$4,933	$6,700	
Art	$966	$1,006	$1,413	
Binders	$12,412	$4,286	$13,728	
Bookcases	$5,062	$1,940	$7,147	
Chairs	$11,285	$7,768	$20,832	
Copiers	$3,960		$22,590	
Envelopes	$750	$669	$1,657	
Fasteners	$88	$159	$150	
Furnishings	$3,980	$2,316	$5,068	
Labels	$207	$300	$940	
Machines	$7,215	$8,990	$35,052	
Paper	$2,287	$2,805	$6,218	
Phones	$13,772	$9,000	$26,712	
Storage	$9,374	$6,125	$14,793	

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Highlight Table in Tableau

Scatter Plot: Reveals relationships between two numeric variables.

- Drag one numeric variable onto the "X" shelf and another onto the "Y" shelf.
- You can adjust the size and color of the points to represent additional dimensions.

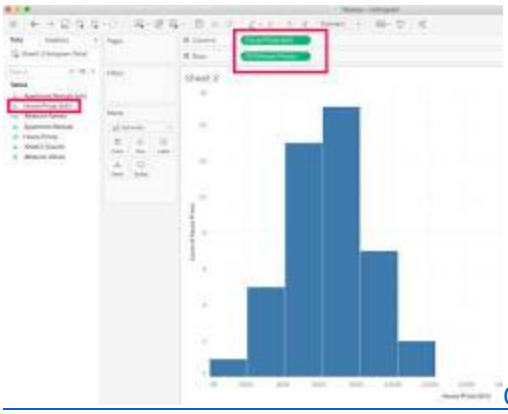


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Scatter Plot in Tableau

Histogram: Shows the distribution of a continuous variable.

- Drag your numeric variable onto the "Marks" shelf and choose "Histogram" as the chart type.
- You can customize the number of bins and adjust the axis labels.

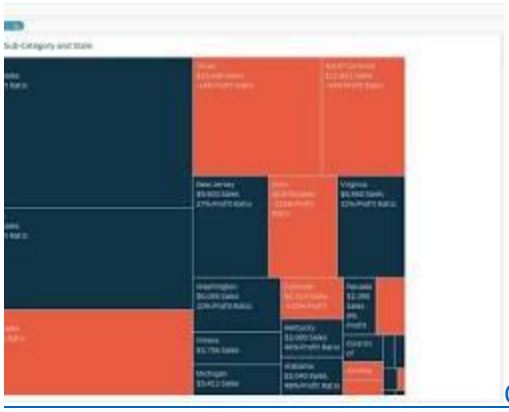


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Histogram in Tableau

Tree Map: Helps visualize hierarchical data and identify areas with the highest or lowest values.

- Drag your dimensions onto the "Color" shelf hierarchically (e.g., region, city, store).
- Drag your measure onto the "Size" shelf.
- You can adjust the color palette and level of detail.

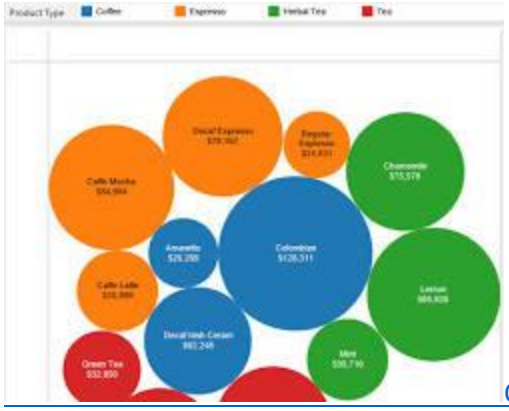


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Tree Map in Tableau

Bubbles: Similar to scatter plots but with additional dimension represented by bubble size.

- Drag your first numeric variable onto the "X" shelf, second numeric variable onto the "Y" shelf, and third numeric variable onto the "Size" shelf.
- You can adjust the color and transparency of the bubbles.

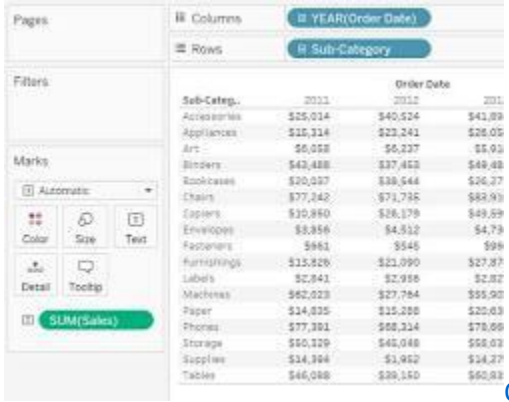


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Bubbles Chart in Tableau

Data Tables: Shows raw data in a tabular format.

- Drag your dimensions and measures onto the "Rows" and "Columns" shelves, respectively.
- You can customize the table by formatting cells, adding sorting, and filtering data.

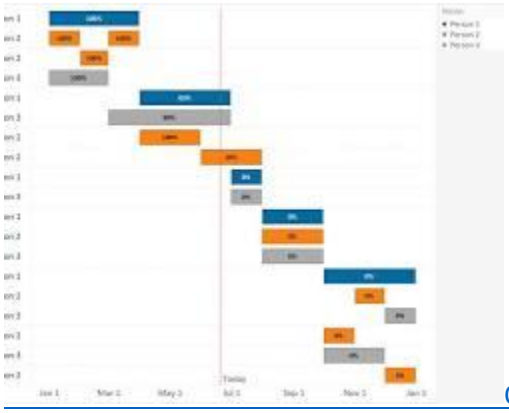


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Data Table in Tableau

Gantt Chart: Visualizes project timelines and tasks.

- Drag your start date dimension onto the "Columns" shelf, end date dimension onto the "Text" shelf, and task name dimension onto the "Rows" shelf.
- Adjust the chart layout and customize the Gantt bars.



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Gantt Chart in Tableau

Box Plots: Show the distribution of a numeric variable across different categories.

- Drag your numeric variable onto the "Marks" shelf and dimension onto the "Columns" shelf.
- Box plots will display quartiles, median, and outliers for each category.

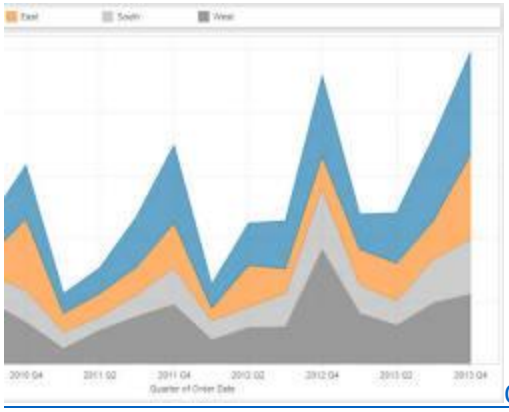


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Box Plot in Tableau

Area Chart: Useful for showing trends over time or comparing multiple measures with overlapping areas.

- Similar to line chart, but drag your measure onto the "Area" shelf instead of "Marks."
- You can adjust the transparency and stacking of the areas.

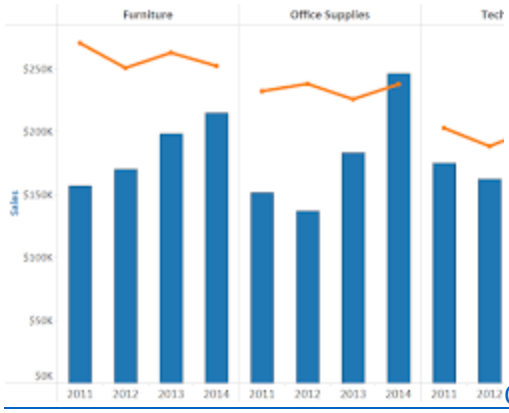


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Area Chart in Tableau

Dual Axis Chart: Combines two different charts on the same canvas to compare trends or relationships.

- Create your desired charts (e.g., line and bar).
- Drag one chart onto the "Marks" shelf and the other onto the "Dual Axis" shelf.
- Adjust the axes and formatting to align the charts effectively.



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Dual Axis Chart in Tableau

Creating Combo Charts in Tableau

Tableau makes it easy to create combo charts. Simply drag and drop different chart types onto the same worksheet. You can then customize the chart by changing the colors, axes, and labels.

Here are some things to keep in mind when creating combo charts:

- Make sure the data types are compatible. Not all data types can be combined in a combo chart. For example, you cannot combine a bar chart with a scatter plot.
- Choose the right chart types. The best chart types for a combo chart will depend on your data and the insights you want to communicate.
- Use clear labels and legends. Make sure your combo chart is easy to understand by adding clear labels and legends.

3.2. Create dashboards and stories

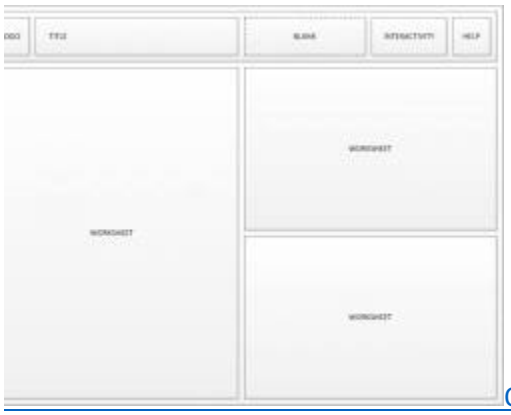
Combining Sheets into a Dashboard using Containers and Layout Options in Tableau

Dashboards in Tableau allow you to combine multiple visualizations (sheets) onto a single canvas to tell a compelling data story. Containers and layout options provide powerful tools for arranging these sheets in an organized and informative way.

Containers:

There are three main container types in Tableau:

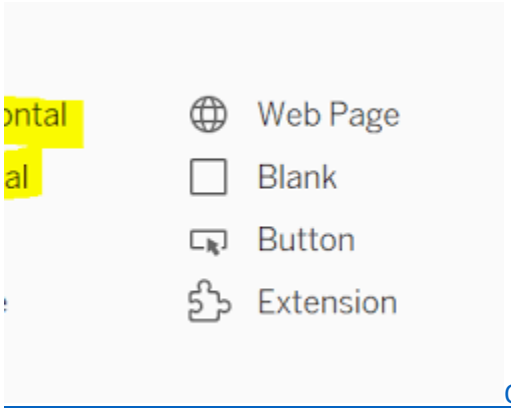
- Horizontal: Arranges sheets side-by-side.



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Horizontal container in Tableau dashboard

- Vertical: Arranges sheets one above the other.



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Vertical container in Tableau dashboard

- Grid: Arranges sheets in a grid-like layout.



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Grid container in Tableau dashboard

You can combine different container types to create more complex layouts. For example, you could use a vertical container to group two related sheets, then place that container within a horizontal container alongside another sheet.

Layout Options:

Within each container, you can further refine the layout using options like:

- Alignment: Align sheets to the left, right, center, or top/bottom of the container.
- Padding: Add space between the container edges and the sheets within it.
- Margins: Add space between the container itself and other elements on the dashboard.

Adding Objects:

Dashboards aren't limited to just sheets. You can also add various objects to enhance your data story, such as:

- Text boxes: Add titles, labels, or annotations to explain your visualizations.
- Images: Include logos, charts, or other visuals to provide context.
- Shapes: Draw lines, circles, or other shapes to highlight specific data points.
- Web objects: Embed webpages or other external content for additional information.

Creating Stories:

Stories in Tableau let you combine multiple dashboards into a sequential presentation, guiding viewers through your analysis step-by-step. This is a great way to showcase complex data insights in a clear and engaging way.

Here's a brief overview of the story creation process:

1. Organize your dashboards: Arrange your dashboards in the desired order for the story.
2. Add titles and captions: Provide context for each dashboard in the story.
3. Set navigation: Define how viewers move between dashboards (linearly, by clicking, etc.).
4. Customize transitions: Add animations or effects between dashboards for a polished look.

3.3. Add interactivity to dashboards

Apply a filter to a view

- Purpose: To focus the view on specific data subsets, enabling users to explore relevant information and uncover insights.
- Methods:
    - Drag-and-drop fields: Directly drag dimensions or measures onto the Filters shelf.
    - Right-click menu: Right-click a field and select "Filter" to explore various filtering options.
    - Filter dialog box: Access advanced filtering options (e.g., custom filters, conditional filters) through the Filters shelf.
- Key considerations:
    - Filter types (context, dimension, measure)
    - Filter hierarchy and context
    - Performance implications of filtering large datasets

Add filter, URL, and highlight actions

- Purpose: To create interactivity and link different views or external resources, enhancing user exploration and storytelling.
- Types of actions:
    - Filter actions: Apply a filter to another sheet based on selections in the current view.
    - URL actions: Open a web page based on selections.
    - Highlight actions: Visually emphasize related data points in other views.
- Implementation: Access action options through the context menu (right-click) on a field or mark.
- Best practices:
    - Use clear and descriptive action names.
    - Test actions thoroughly to ensure expected behavior.

Swap sheets by using parameters or sheet selector

- Purpose: To dynamically change displayed sheets within a dashboard, offering flexibility and customization for users.
- Methods:
    - Parameters: Create a parameter with values corresponding to sheet names, then use it to control sheet visibility.

    - Sheet selector: Add a built-in sheet selector object to a dashboard for user-driven sheet switching.

- Best practices:
    - Use parameters for conditional or calculated sheet swapping.
    - Employ sheet selectors for straightforward user-controlled switching.

Add navigation buttons

- Purpose: To provide intuitive navigation within dashboards, guiding users through content and improving usability.
- Implementation:
    - Create custom buttons using shapes and images.
    - Assign actions (e.g., Go to Sheet, Go to URL) to buttons.
- Best practices:
    - Use clear and consistent button styles.
    - Position buttons logically for easy access.
    - Provide visual cues for button states (hover, active).

Implement user guiding sentences (click…, hover…, menu options)

- Purpose: To offer clear instructions and prompts within the dashboard, aiding user interactions and understanding.
- Methods:
    - Use text boxes or annotations to display guiding sentences.
    - Conditionally show or hide guidance based on user actions.
- Best practices:
    - Use concise and actionable language.
    - Test guidance with diverse users to ensure clarity.
    - Consider accessibility for users with visual impairments.

3.4. Format dashboards

Apply color, font, shapes, styling:

- Colors: Use color palettes strategically to highlight key elements, distinguish data points, and evoke emotions. Understand color theory and best practices for accessibility.
- Fonts: Choose readable fonts that complement your data and design. Apply consistent text formatting across your dashboard.

- Shapes: Go beyond circles and squares! Utilize custom shapes to symbolize specific data points or categories, adding visual interest and clarity.
- Styling: Leverage features like borders, transparency, and sizing to make specific elements stand out or blend seamlessly. Master data formatting options to refine how values are displayed.

Add custom shapes and color palettes:

- Custom Shapes: Step outside the box! Tableau allows you to import or create custom shapes to represent unique data points or add a distinct brand identity to your visualizations.
- Color Palettes: Craft bespoke color palettes tailored to your data and brand. This empowers you to tell a specific story or evoke desired emotions through color choices.

Add annotations:

- Text Annotations: Provide additional context or highlight specific data points directly on your visualizations using text labels.
- Shapes and Images: Enhance your storytelling by adding custom shapes or images as annotations to draw attention to specific areas or trends.

Add tooltips:

- Hover Details: Reveal deeper insights when users hover over data points. Use tooltips to display additional data values, metrics, or explanations.
- Interactive Storytelling: Link tooltips to other dashboards or web pages to create layered explorations and guide users through your analysis.

Mastering these techniques will elevate your Tableau dashboards from basic visualizations to compelling data stories. Remember:

- Consistency is key: Maintain a consistent visual style throughout your dashboard for professional and organized look.
- Clarity above all: Prioritize user understanding. Use design elements to enhance, not obscure, your data insights.
- Experiment and have fun! Explore different options and find creative ways to showcase your data effectively.

Apply padding:

- Padding adds blank space around the edges of your chart elements, like axis labels, titles, and legend entries. This improves readability and prevents elements from feeling cramped.
- You can control padding individually for elements or globally for the entire chart. Tableau offers various padding options to customize the spacing exactly as you need.

Remove gridlines, row-level and column-level bands, and shading:

- Gridlines are horizontal and vertical lines across the chart area. They can be helpful for reference, but sometimes clutter the visualization and distract from the key insights. You can completely disable them or selectively modify their appearance.
- Row-level and column-level bands add color stripes based on data categories. While they can help identify groups, overuse can overwhelm the visual. Removing them can simplify the view and emphasize trends or outliers.
- Shading applies different shades of a color to data points based on their values. While highlighting data variation, it can obscure details in dense visuals. Removing shading can make the data points clearer.

Apply responsive design for specific device layouts:

- Responsive design adapts your visualization to different screen sizes and devices. This ensures optimal viewing and interaction regardless of whether someone views it on a desktop, tablet, or mobile phone.
- Tableau offers built-in layout containers and conditional formatting options to create different configurations for various device types. You can adjust aspects like chart size, font sizes, axis placement, and element visibility based on screen dimensions.

Mastering these techniques allows you to:

- Enhance the clarity and focus of your visualizations.
- Create aesthetically pleasing dashboards that users find engaging.
- Ensure your data insights are accessible and readily interpretable on any device.

Domain 4: Publish and Manage Content on Tableau Server and Tableau Cloud

4.1. Publish Content

Publish a workbook:

- This allows you to upload your entire Tableau workbook, including dashboards, worksheets, and data sources, to Tableau Server or Tableau Online.
- This makes your work accessible to others within your organization who have the appropriate permissions.
- There are various publishing options based on your needs, such as embedding specific dashboards in webpages or sharing privately with specific users.

Publish a data source:

- This lets you share the underlying data used in your visualizations without sharing the entire workbook.
- This can be helpful if others need access to the data for their own analysis or if you want to keep your workbook size smaller.
- You can publish data sources from various formats, including Excel, CSV, and databases.

Print content:

- This enables you to create high-quality printable versions of your dashboards and worksheets.
- You can adjust layout, format, and size for optimal printing.
- This is useful for sharing insights with people who may not have access to Tableau or for creating physical reference materials.

Export content:

- This allows you to export your visualizations in various formats for further use outside of Tableau.
- Common export formats include PDF, image files (PNG, JPEG), and Excel.
- This is useful for incorporating your visuals into presentations, reports, or other documents.

Additional Considerations:

- When publishing or exporting, you'll need to consider security settings and ensure only authorized users have access to your data.
- You can schedule automatic refreshes for published data sources to keep your visualizations up-to-date.
- Learning best practices for formatting and layout will enhance the effectiveness of your published and exported content.

4.2. Schedule data updates

Scheduling Data Extract Refreshes in Tableau (4.2.1)

Data extracts are snapshots of your underlying data sources. Keeping them current is crucial for reliable analysis. Tableau offers several ways to schedule refreshes:

1. Manual Refresh: Click the "Refresh" button in the Data window. This is quick for small datasets but impractical for large datasets or recurring updates.

2. Extract Refresh Schedule: Set a refresh schedule within the data source in Tableau Desktop. You can choose specific dates/times, intervals (hourly, daily, etc.), or trigger refreshes based on workbook events like opening or filtering.

3. Tableau Server/Online Schedule: If you publish your workbook to Tableau Server or Online, you can use its "Schedules" functionality to manage refreshes across multiple workbooks. This offers more flexibility and centralized control.

4. Tableau Prep (Scheduled Flows): For advanced data prep before extracting, you can schedule your Tableau Prep workflow to run automatically. This ensures pre-processed data is always ready for analysis when the extract refreshes.

Tips:

- Consider refresh frequency based on data volatility and user needs.
- Monitor refresh history for errors and adjust schedules as needed.

- Utilize alerts to notify stakeholders about successful or failed refreshes.

Scheduling a Tableau Prep Workflow (4.2.2)

Tableau Prep automates data cleaning and transformation tasks. Scheduled workflows ensure your analysis is based on consistently prepared data.

How to schedule a workflow:

1. Open your workflow in Tableau Prep.
2. Click the "Schedule" button in the toolbar.
3. Choose a schedule type:
    - Run Once: Execute the workflow at a specific date/time.
    - Recurring: Set a regular schedule (daily, hourly, etc.).
    - Trigger: Run the workflow when a specific event occurs (file arrival, database update, etc.).
4. Configure additional settings like output destination and notifications.
5. Start the schedule.

Benefits of scheduling Prep workflows:

- Eliminates manual data prep tasks.
- Ensures data consistency and quality.
- Improves data analysis efficiency.
- Enables self-service data access for non-technical users.

4.3. Manage Published workbooks

Creating Alerts in Tableau

An alert in Tableau is a notification triggered when a specific condition met within your data visualization. This helps you stay informed about critical changes or trends without constantly monitoring your dashboards. There are two main types of alerts in Tableau:

- Data alerts: These are based on data values within your visualization. You can set a threshold for a specific metric, and the alert will fire if the value crosses above or below that threshold. For example, you could set an alert for when sales fall below 10% of the average.
- Viewpoint alerts: These are triggered when specific user interactions with your dashboard occur. For example, you could set an alert for when someone filters the data to a specific time period or selects a particular mark.

Creating an alert involves:

1. Selecting the visualization: Choose the dashboard or workbook where you want to set the alert.

2. Defining the condition: Specify the metric, threshold, and timeframe for the data alert, or the specific interaction for the viewpoint alert.
3. Configuring the notification: Choose how you want to be notified when the alert triggers. This can be through email, SMS, or integration with other platforms.
4. Sharing and scheduling: Decide whether to share the alert with other users and set a schedule for when it should be active.

Creating Subscriptions in Tableau

A subscription in Tableau allows you to automatically send your visualizations or data extracts to specific recipients at regular intervals. This keeps relevant stakeholders informed without them needing to manually access the dashboards themselves.

Creating a subscription involves:

1. Selecting the content: Choose the dashboard, workbook, or data extract you want to share.
2. Defining the schedule: Specify how often the content should be sent (e.g., daily, weekly, monthly) and at what time.
3. Adding recipients: Choose the email addresses or groups who should receive the subscription.
4. Formatting the email: Customize the email subject line, message body, and attachment format (e.g., PDF, image).
5. Sending a test: Send a dummy email to ensure everything is working correctly before scheduling the regular subscription.

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